

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Civil) (Credit System) Degree Examinations**

May - June 2023

20CVE502 - ADVANCED CONCRETE TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of **IS10262-2019** is permitted.

Unit – I

Marks BT* CO* PO*

- | | | | | | |
|-------|---|----|-----|---|---|
| 1. a) | Calculate the percentage of C_3S (tricalcium silicate) in a cement sample, given that the X-ray diffraction (XRD) pattern shows peaks at 2θ angles of 28.2° , 47.6° , and 61.5° , corresponding to the (100), (021), and (002) planes, respectively. | 10 | L*3 | 1 | 2 |
| b) | What factors can affect the accuracy of measurements taken with a Rebound hammer, and how can they be controlled? | 10 | L3 | 2 | 1 |
| 2. a) | A civil engineer is tasked with designing a concrete mix with a target compressive strength of 30 MPa and a maximum porosity of 10%. The engineer has access to two types of aggregates: crushed granite and rounded river gravel. The water-cement ratio is fixed at 0.5. The engineer conducts two tests: one with 100% crushed granite and another with 50% crushed granite and 50% river gravel. The results are as follows:
100% crushed granite: Compressive strength = 35 MPa, Porosity = 8%
50% crushed granite + 50% river gravel: Compressive strength = 28 MPa, Porosity = 7%
Which mix should the engineer use based on the criteria of compressive strength and porosity? | 10 | L3 | 1 | 2 |
| b) | Interpret the results of an elastic modulus test on concrete. What information can be gleaned from the modulus measurement? How does it relate to the strength and durability of the concrete? | 10 | L4 | 2 | 1 |
| 3. a) | How does the Interfacial Transition Zone impact the performance of concrete? | 10 | L3 | 1 | 4 |
| b) | How do variations in end conditions, capping, H/D ratio, rate of loading, and moisture condition affect the reliability of concrete strength and elasticity test results? | 10 | L3 | 2 | 4 |

Unit – II

- | | | | | | |
|-------|---|----|----|---|---|
| 4. a) | What factors can contribute to autogenous shrinkage in concrete, and how does it differ from drying shrinkage? | 10 | L3 | 3 | 3 |
| b) | How do alkali-activated binders compare to traditional Portland cement in terms of their environmental impact and sustainability? | 10 | L3 | 4 | 7 |
| 5. a) | How does the presence of chloride ions accelerate the corrosion of steel reinforcement in concrete? | 10 | L3 | 3 | 2 |
| b) | How does the addition of encapsulated healing agents affect the properties of self-healing concrete? | 10 | L3 | 4 | 2 |
| 6. a) | How does carbonation affect the pH of concrete, and what impact does this have on its durability? | 10 | L3 | 3 | 6 |

P.T.O.

- b) What challenges might arise in using carrot extract as a cement additive, and how can these challenges be addressed in the production process? Consider factors such as availability, cost, and potential effects on the mechanical properties of the cement.

10 L3 4 7

Unit – III

7. a) How do admixtures such as superplasticizers and air-entraining agents affect the properties of a concrete mix, and what are their applications in different scenarios?
- b) How does the selection of cement type (e.g. Portland cement vs. blended cement) affect the properties of a concrete mix, and what are the implications for cost and sustainability?
8. Design M 80 Concrete for a slump of 100mm, assume all necessary data suitably.

10 L3 5 1

10 L3 5 7

20 L4 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
